Brazilian E-Commerce Insights

Project Caravela — Data Pipeline Findings

Analysis Period: January 2017 – August 2018 | ~100,000 Orders | R$15.8M GMV

Executive Summary

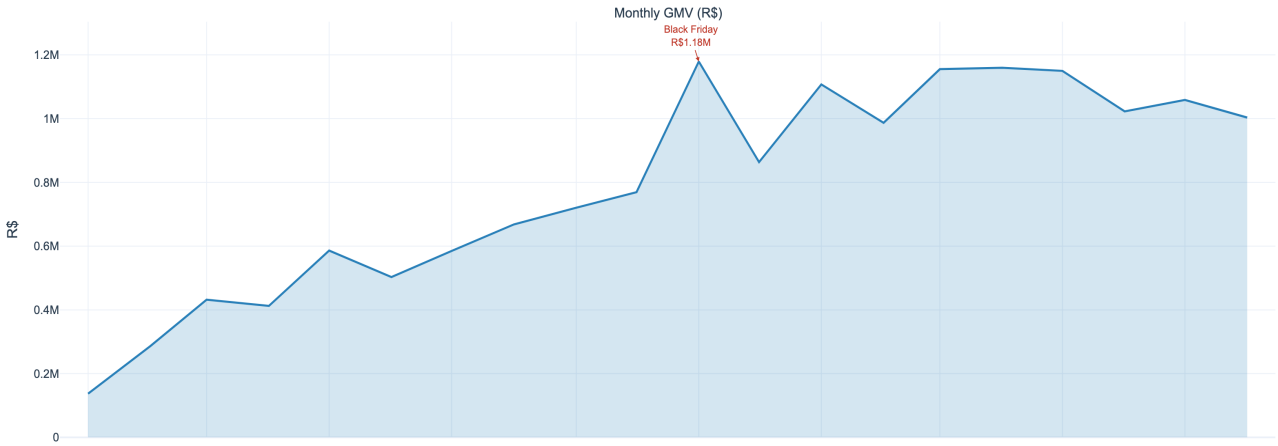
R$15.8M	98,353	63.6	91.9%
Total GMV	Total Orders	NPS Score	On-Time Rate

R$160.51	3.1%	0.46%	~103%
Avg Order Value	Repeat Purchase Rate	Cancellation Rate	H1→H2 2017 Growth

- GMV doubled from H1 to H2 2017, driven by volume (not higher order values)
- 96.9% of customers make only one purchase — retention is the critical challenge
- Northeast delivery lags at 85.7% on-time (vs 92.9% South) — impacting satisfaction
- Southeast dominates with 64.6% of GMV; São Paulo alone = 37.4%

Market Overview: Monthly GMV & Volume

Jan 2017 – Aug 2018 | Peak: Nov 2017 (Black Friday)



H1→H2 2017: ~103% GMV growth

H2 2017→H1 2018: ~38% (decelerating)

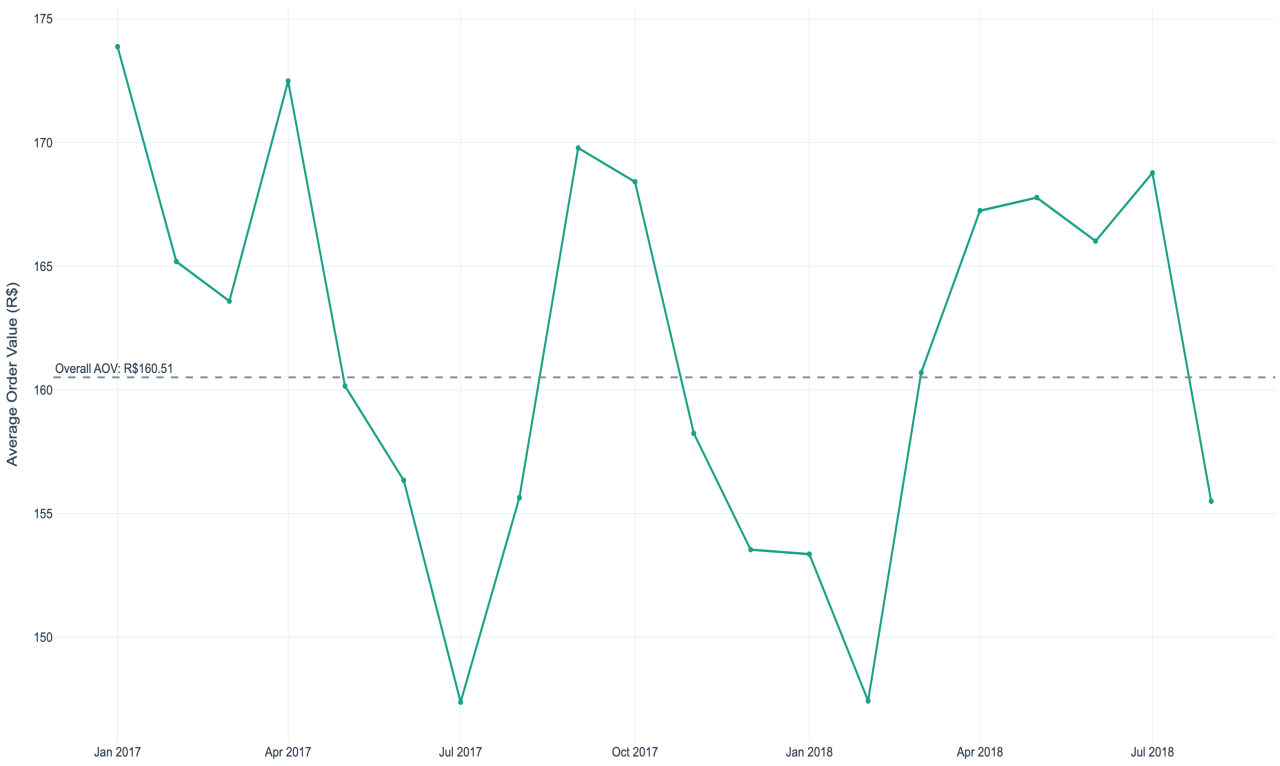
Nov 2017 peak: 7,451 orders, R$1.18M

Black Friday was volume-driven — AOV dropped 6%

Sep/Oct 2018 excluded (data artefacts)

Average Order Value Trend

Stable at R$160.51 — growth is volume-driven, not ticket-size-driven



Overall AOV: R$160.51

Remarkably flat across 20 months

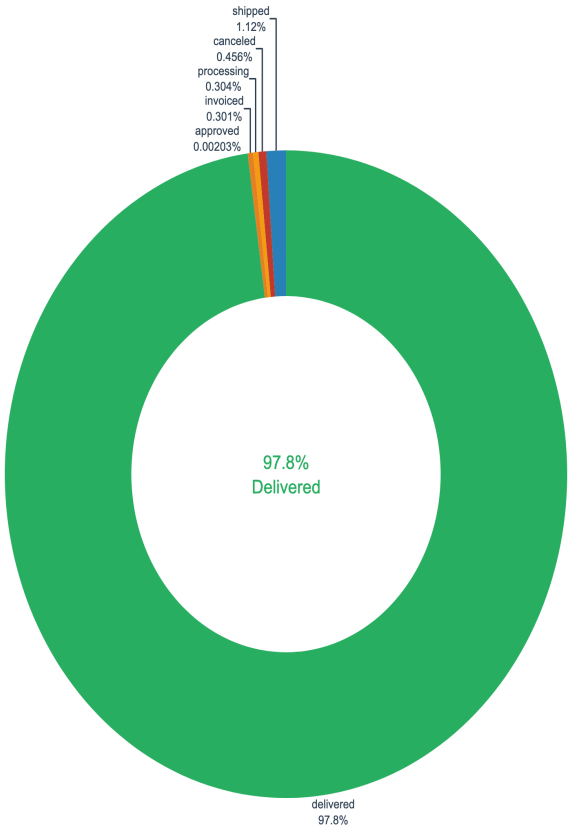
Credit card AOV: R$152

Boleto AOV: R$127 (−20%)

Instalment access enables larger baskets

Order Fulfilment & Cancellation

97.8% delivered | 0.46% cancellation rate



97.8% of orders delivered successfully

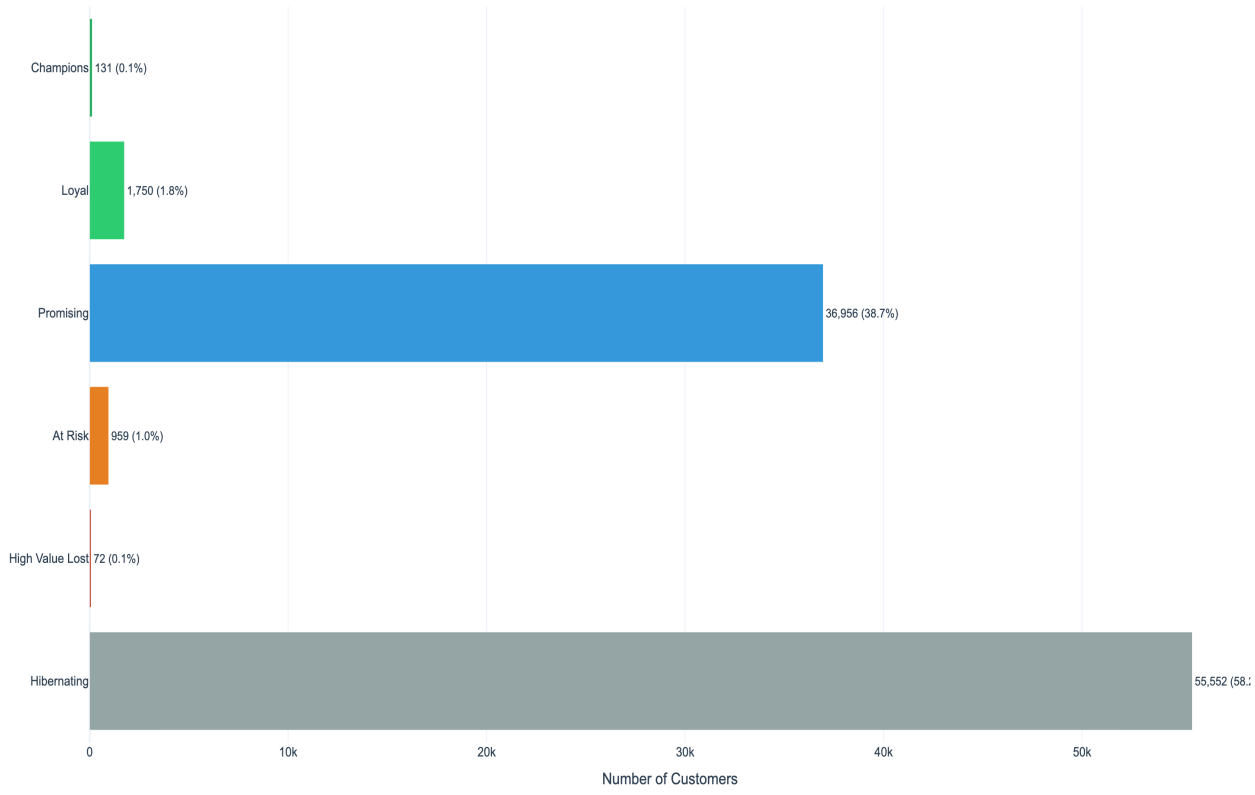
Cancellation rate: just 0.46% (448 orders)

Strong inventory management

Remaining 1.7%: in-transit pipeline statuses

Customer Segmentation (RFM Analysis)

95,420 unique customers | Reference date: 31 Aug 2018



58.2% Hibernating (55,552)

38.7% Promising (36,956)

Only 3.1% repeat purchase rate

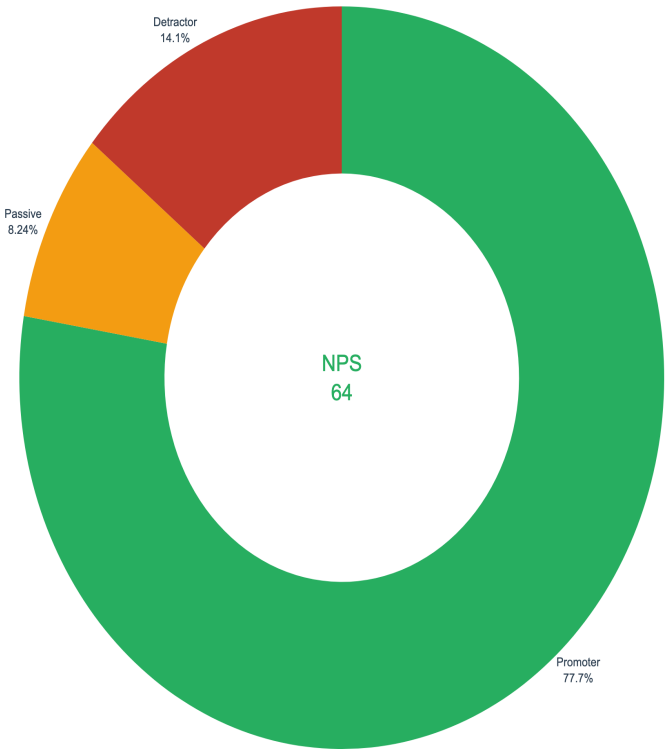
131 Champions (0.1%) — crown jewels

Marketplace model = low platform loyalty

Promising segment = top retention target

Customer Satisfaction — NPS Proxy

Net Promoter Score: 63.6 (Strong)



Promoters (score 4–5): 77.7%

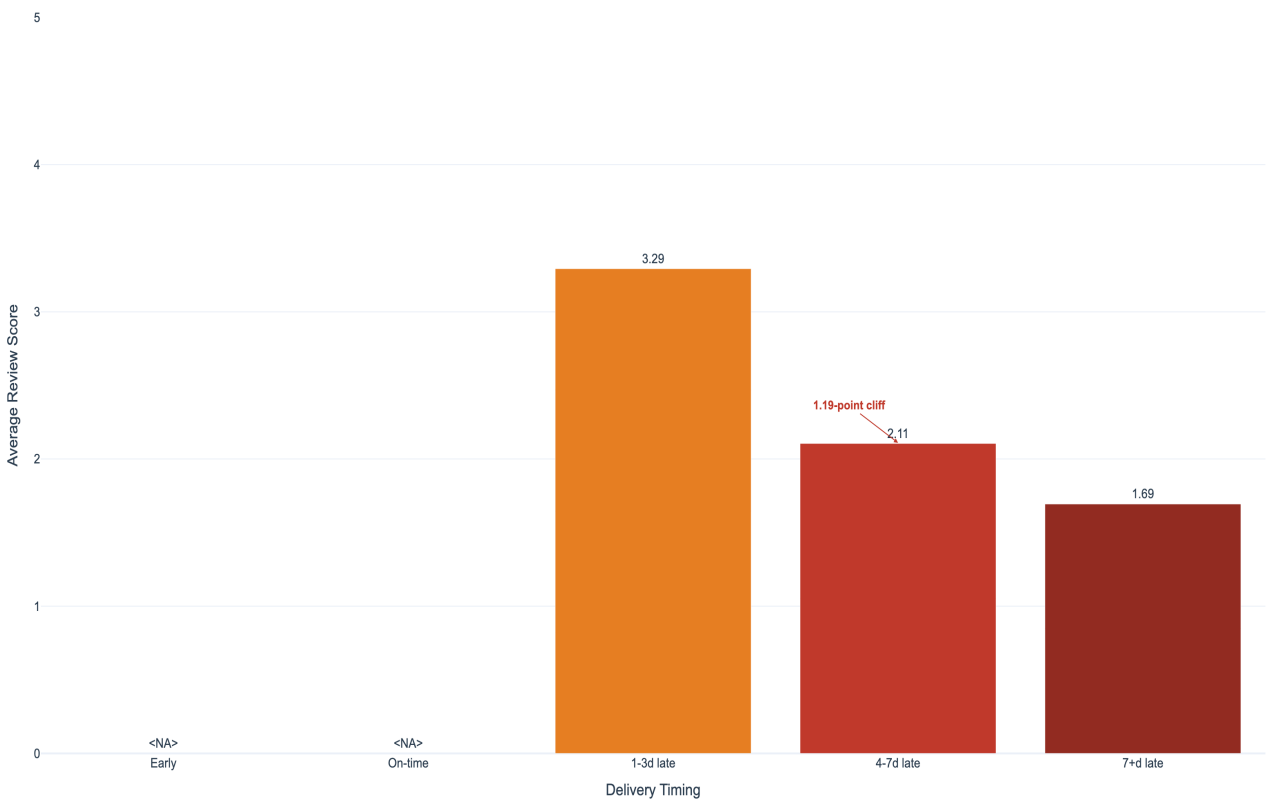
Detractors (score 1–2): 14.1%

NPS = 77.7% − 14.1% = 63.6

Strong overall, but delivery delays are the #1 driver of detractors

Key Insight: Delivery Delays Destroy Satisfaction

2.61-point score drop from early delivery to 7+ days late



Early delivery: score 4.30

On-time: 3.89

1–3 days late: 3.29

4–7 days late: 2.11 (−1.19 cliff!)

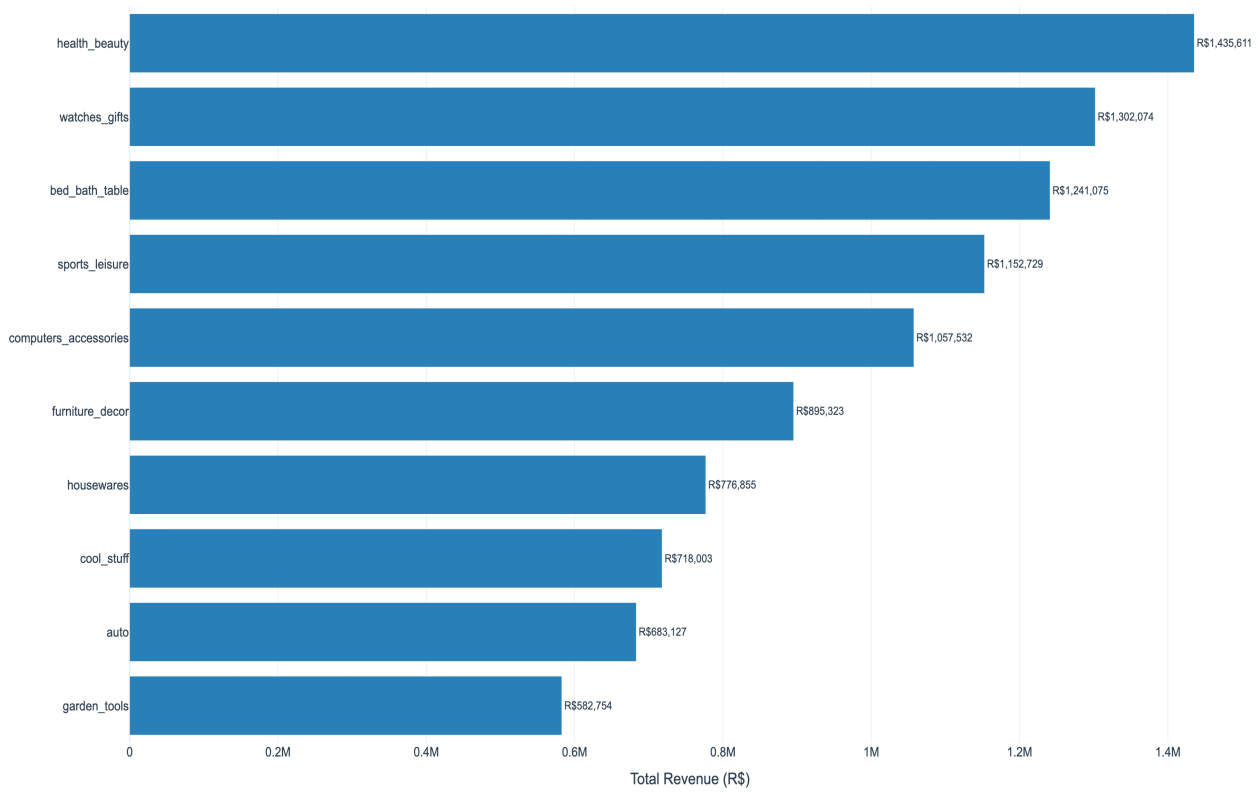
7+ days late: 1.69

The 4-day mark is where customer patience breaks.

6,359 orders (6.6%) were late.

Product & Revenue: Top Categories

Top 15 categories = 76.4% of GMV | 74 categories total



#1 Health & Beauty: 9.1%

#2 Watches & Gifts: 8.2%

#3 Bed, Bath & Table: 7.9%

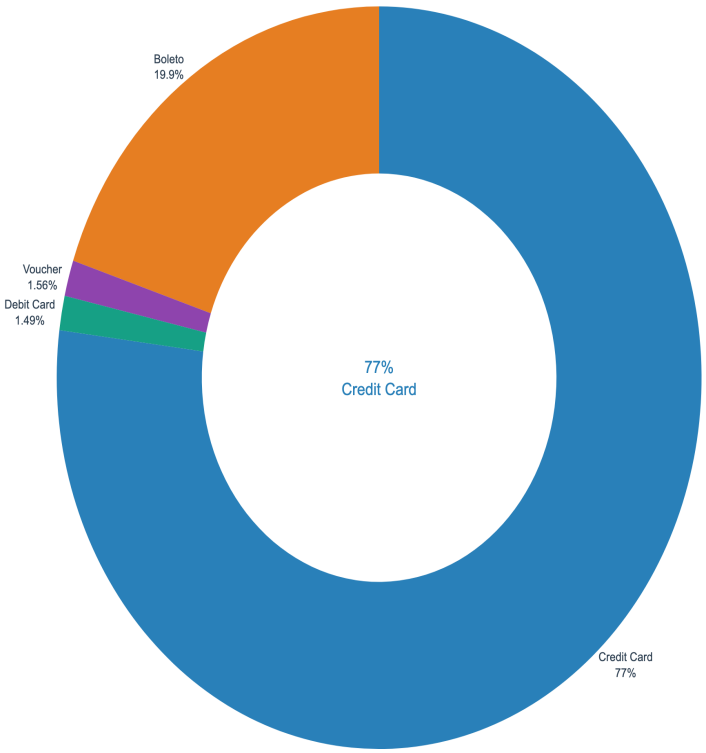
Category Gini: 0.71 (long tail)

HHI: 484 (competitive, no dominance)

Freight = 14.2% of GMV (R$2.24M)

Payment Method Distribution

Credit card dominance with instalment culture



Credit Card: 77.0% of orders

Boleto: 19.9%

Debit: 1.5% | Voucher: 1.6%

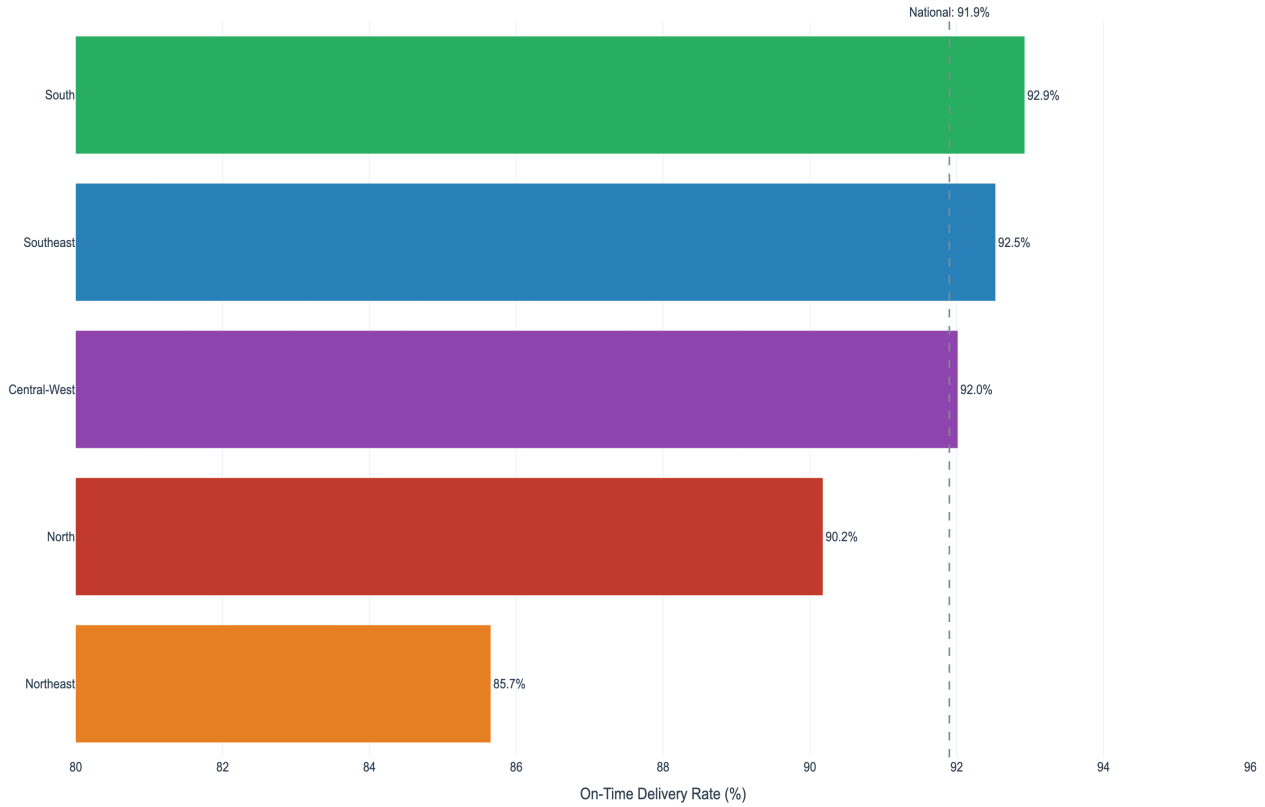
Median instalments: 3

7.3% choose 10+ instalments

CC AOV 20% higher than boleto

Delivery Performance by Region

National on-time rate: 91.9% | Northeast lags at 85.7%



South: 92.9% (best)

Southeast: 92.5%

Central-West: 92.0%

North: 90.2%

Northeast: 85.7% (worst)

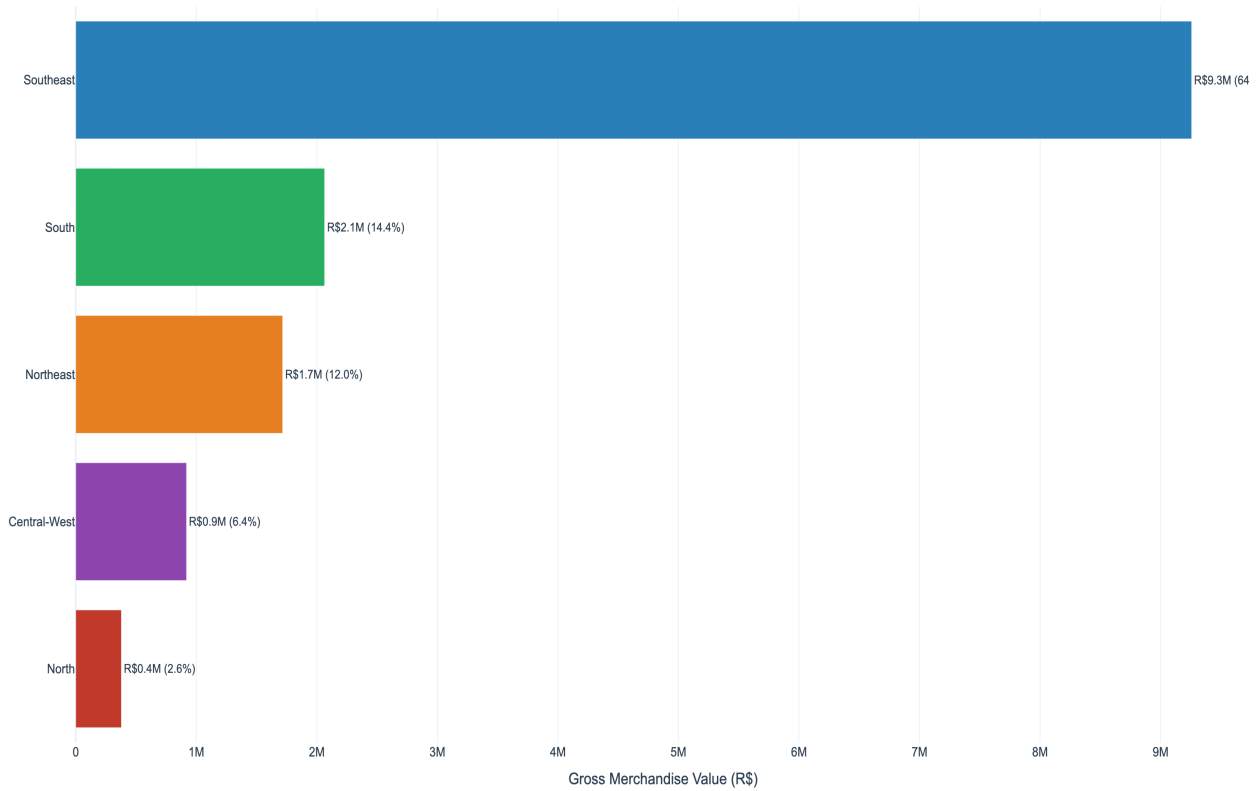
NE transit: 20.6 days avg

vs SE: 12.9 days avg

7.2pp gap → satisfaction gap

Geographic Revenue Distribution

Southeast = 64.6% of GMV | São Paulo alone = 37.4%



Southeast: R$10.2M (64.6%)

South: R$2.3M (14.5%)

Northeast: R$1.9M (11.9%)

Central-West: R$1.0M (6.5%)

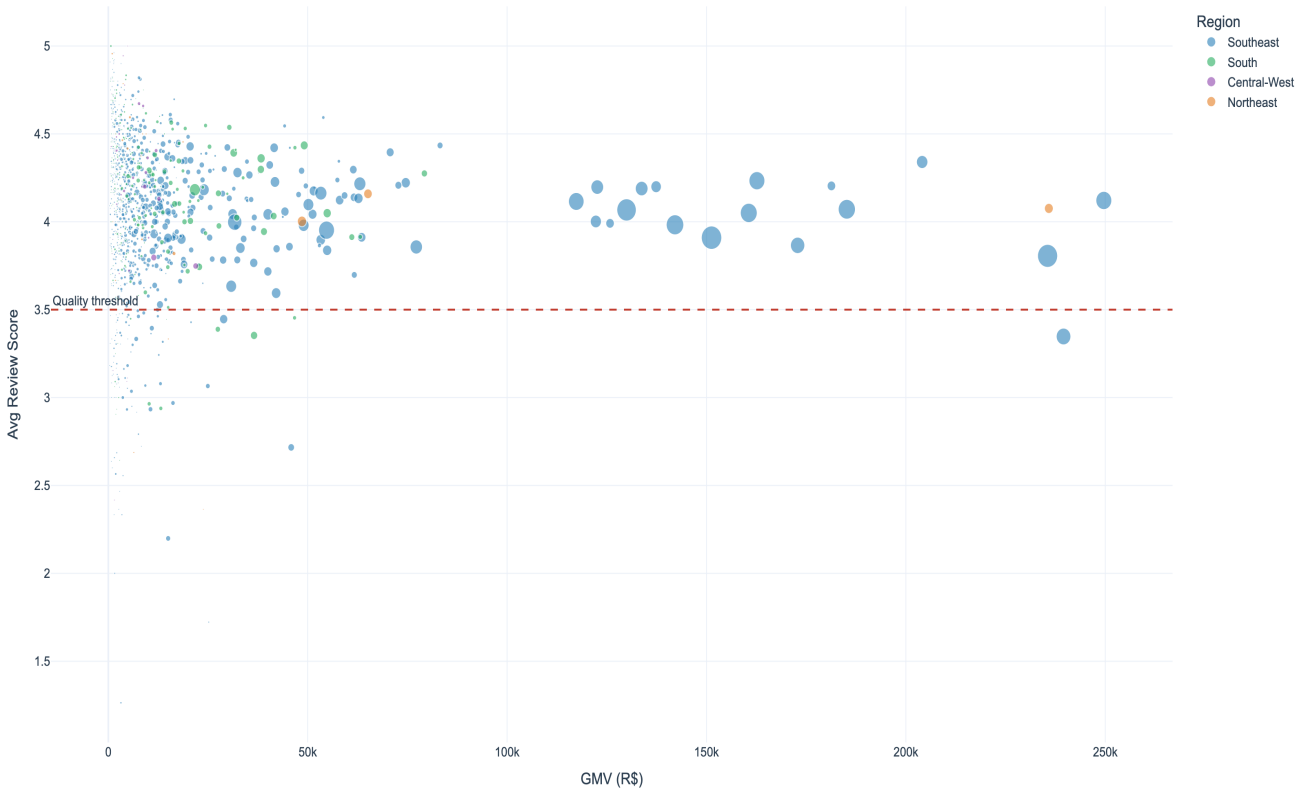
North: R$0.4M (2.6%)

North + Central-West = 9.1%

but >20% of Brazil's population

Seller Performance Landscape

3,068 sellers | Top 18.3% generate 80% of GMV



Seller Gini: 0.78 (high inequality)

CR4: 6.1% (no monopoly)

HHI: 35 (competitive market)

767 Premium sellers (score ≥4.3)

37 At Risk sellers (score <3.0)

Avg platform score: 3.98/5

Key Risks & Mitigations

Technical Risk: Historical Data Limitations

- Dataset covers 2016–2018 — consumer behaviour has evolved since then
- Mitigation: Pipeline is repeatable; refresh with current data and monitor metric drift quarterly

Business Risk: Single-Purchase Customer Dominance

- 96.9% of customers buy only once — CAC amortised over a single transaction
- As high-penetration regions saturate, acquisition costs increase unsustainably
- Mitigation: Target 36,956 Promising customers with post-purchase engagement programmes

Business Risk: Regional Delivery Disparity

- Northeast on-time rate 85.7% vs South 92.9% — a 7.2pp gap
- Late deliveries score 1.69 vs 4.30 for early — compounding NE growth barriers
- Mitigation: Regional carrier partnerships + distribution centres in NE and North

Strategic Recommendations

1. Customer Retention Programme

- Target 36,956 Promising segment customers within 30 days of first purchase
- Evidence: Repeat rate declining from 5.3% (2017) to 2.6% (2018 cohorts)

2. North & Northeast Delivery Infrastructure

- Partner with regional carriers; establish fulfilment hubs in Recife and Manaus
- Evidence: NE on-time rate 85.7% vs South 92.9%; North transit 23.6 days avg

3. Optimise Delivery Promise Accuracy

- A/B test tighter estimates in SE (actual 12.9d vs promised 24.5d = 11.6d buffer)
- Hypothesis: Tighter promises increase checkout conversion without hurting on-time rate

4. Seller Quality Intervention

- 37 At Risk sellers (avg score 2.74, 5.4% cancel rate) — 90-day improvement window

5. Instalment-Driven AOV Uplift

- CC AOV R$152 vs boleto R$127; promote instalments to boleto-inclined buyers

Thank You

Project Caravela | Questions & Discussion

Interactive dashboard available at: streamlit run dashboard.py